Dong et al. 2020 — Prostate Cancer

Cell-state manifold analysis · 10x Genomics · 9,698 malignant cells (4 samples) · CCCA meta-atlas

GO / NO-GO GATES

Gate 1 — mixing + continuity **NO-GO**

Malignant cells split into two disconnected lineage lobes — adenocarcinoma (patients #1/#2/#5) and small-cell neuroendocrine NEPC (patient #6) — and never sample-integrate. Judged on the regenerated embedding across a theta sweep with per-neighborhood iLISI (max 4): iLISI stays ~1.2 and each lobe is a per-patient archipelago; n_pcs-invariant (20/30/40). Mixing appears only at theta=8 by destroying the embedding. Not mixed, not continuous -> do not force a trajectory (01_task.md sec.2, HIGH).

Gate 2 — module geography **NA**

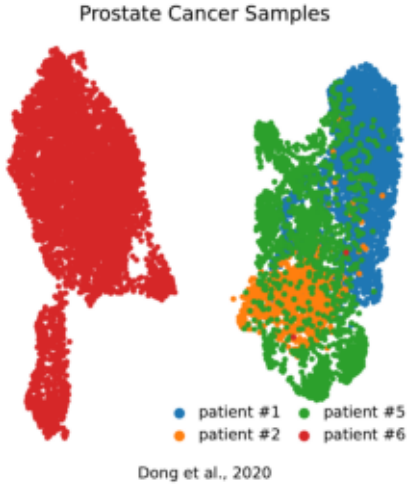
Not evaluated: the gate chain stops at the Gate 1 NO-GO (no manifold/trajectory is fit).

Gate 3 — hub cross-repression **NA**

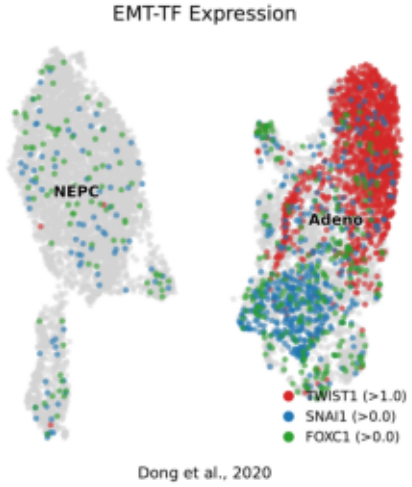
Not evaluated (branching-only; requires a fitted tree, which Gate 1 precludes).

STEP FIGURES

01 · Samples (Gate 1)



02 · EMT-TF expression



Dataset. Prostate cancer (Dong et al. 2020), 10x; 6 patients, 21,292 cells total; 9,698 malignant, of which 4,933 adenocarcinoma and 4,765 small-cell NEPC. Patients #3/#4 contribute no malignant cells.

Verdict. The embryonic-program pattern does NOT reproduce. The malignant compartment is two distinct prostate-cancer lineages (adeno vs NEPC) that neither integrate nor form one continuous manifold; the adeno lobe alone is a per-patient archipelago. Gate 1 NO-GO — no trajectory fit. EMT-TF program (TWIST1 dominant) is confined to the adeno lobe.

PARAMETER CHOICES

Min-cell threshold:	>=100 malignant/sample -> keeps all 4 present samples (no low-count tail; #3/#4 have 0)
Overdispersed genes:	1,670 (scFates GAM fit)
Integration:	Harmony on sample; theta sweep 0/2/5/8; n_pcs=20 (scree tail; grid-invariant 20/30/40)
Gate-1 metric:	per-neighborhood iLISI (k=90) + embedding read; mean iLISI 1.0/1.2/1.6/1.9 by theta (max 4)
Lobe biology:	adeno: Gast-pSom r=+0.55, both oppose Crest (reproduces ref); NEPC: distinct AR+/CHGA+ lineage
Annotation:	Adeno / NEPC — data markers (AR,NKX3-1,FOLH1 vs CHGA,SYP) + source-reported histology
Geometry:	none fit (Gate 1 NO-GO)
Root node:	n/a (no trajectory)
EMT-TFs shown:	TWIST1 (dominant), SNAI1, FOXC1 — detection-gated; VIM excluded (not a TF)